INCH-POUND

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23 January 2024
SUPERSEDING
A-A-59474C
10 January 2013

COMMERCIAL ITEM DESCRIPTION

INSULATION TAPE, ELECTRICAL; HIGH TEMPERATURE, POLYTETRAFLUOROETHYLENE, PRESSURE-SENSITIVE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for polytetrafluoroethylene, high-temperature insulating, pressure-sensitive, adhesive, and electrical tapes.

2. CLASSIFICATION. The electrical tape shall conform to the following types and classes:

2.1 Type.

Type I - Smooth backing (see 7.1.1).
Type II - Treated backing (for insulation varnish) (see 7.1.2).

2.2 Classes.

Class	Backing thickness (mils)
1	1.5 to 2.5
2	2.6 to 3.5
3	3.6 to 4.5
4	4.6 to 5.5
5	5.6 to 6.5
6	6.6 to 7.5

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3. SALIENT CHARACTERISTICS

3.1 Backing material. Backing material shall be fused polytetrafluoroethylene resin. Reprocessed material shall not be used in the manufacture of the backing material. Type II backing shall be treated to accommodate an insulation varnish.

3.1.1 Backing thickness. The backing thickness shall be specified (see 7.3) and shall be not less than 1.5 mils (0.038 mm), exclusive of the adhesive thickness when tested in accordance with ASTM D3652/D3652M.

3.1.2 Tensile strength and dielectric breakdown. The tensile strength and dielectric breakdown of the backing material shall meet the requirements in table I for the backing thickness specified when tested in accordance with ASTM D3759/D3759M and 5.3.1.

TABLE I. Backing thickness required tensile strength and dielectric breakdown values.

Backing thickness (ASTM D3652/D3652M)		Tensile strength (minimum) (ASTM D3759/D3759M)		Dielectric breakdown (ASTM D149)	
mils	mm	Type I and Type II lb/in width	N/mm width	Type I (volts/min)	Type II
1.5 - 2.5	0.038 - .065	7.0	1.23	4,200	3,800
2.6 - 3.5	0.066 - .089	12.0	2.10	7,200	6,500
3.6 - 4.5	0.090 - .115	17.0	2.98	10,300	9,000
4.6 - 5.5	0.116 - .141	22.0	3.85	12,700	11,500
5.6 - 6.5	0.142 - .166	27.0	4.73	15,400	14,000
6.6 - 7.5	0.167 - .191	32.0	5.60	18,000	16,500

3.1.3 Melting point. The melting point of the backing material shall be 621 ±18 °F (327 °C ± 10 °C) when tested in accordance with ASTM D4894.

3.1.4 Specific gravity. The specific gravity of the backing material shall be not greater than 2.19 and shall be not less than 2.14 when tested in accordance with ASTM D792, Method A.

3.1.5 Adhesive. The adhesive shall be a pressure-sensitive silicone polymer applied uniformly, free of voids and lumps, to the backing and shall not require heat treatment or any other means of preparation prior to application.

3.2 Rolls. Type I tape shall be uniformly wound on cores, adhesive side in, without liner or separator. Type II shall be uniformly wound on cores, adhesive side in, and is permitted to have a liner.

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3.2.1 Core. The core shall be rigid to prevent distortion of the roll during shipment and use. The core shall have a tolerance of ±1/16 inch (1.6 mm) of the nominal inner diameter specified (see 7.3).

3.2.2 Dimensions. Unless otherwise specified, the tape shall be 36 yards (32.9 m) in length. The width of the tape shall be as specified by the acquiring activity with a tolerance that shall be ± 0.03125 inch (0.8 mm). (see 7.3 and 7.4)

3.2.3 Marking. Unless otherwise specified (see 7.2 and 7.3), each roll shall be marked in or on the edge of the core with the manufacturer's name, designation of the product, specification number, type, class, month, and year of manufacture.

3.2.4 Splices. The tape shall not contain any splices on a 36-yard (32.9 m) roll. For rolls greater than 36 yards in length, the number of splices shall be as specified by the acquiring activity (see 7.3).

3.3 Varnish adhesion (type II only). Insulation varnish shall not crack or peel away from treated backing, and there shall not be any bare spots after air drying of the varnish when tested in accordance with 5.3.2.

3.4 Shelf life. When packaged in accordance with 6.1 and stored under ordinary temperate zone warehouse conditions where temperatures of -15 °C to 50 °C are possible, the tape shall meet all of the requirements of this CID for a minimum of one year after the date of receipt at destination. The shelf life may be extended by the procuring activity (see 7.3) for one-year intervals after reacceptance testing for conformance to adhesion requirements 3.10.

3.5 Elongation. The elongation value between benchmarks 2 inches (51 mm) apart shall be not less than 100 percent when tested in accordance with ASTM D3759/D3759M.

3.6 Moisture absorption. The moisture absorption on the backing material without the adhesive mass shall be not greater than 0.1 percent when tested in accordance with ASTM D570.

3.7 Volume resistivity. Volume resistivity shall be not less than $1x10^{14}$ ohms/cm when tested in accordance with ASTM D257.

3.8 Adhesion as received. The adhesion as received shall not be less than 16 ounces/inch (0.175 N/mm) when tested in accordance with ASTM D3330/D3330M.

3.9 Resistance to curl and twist. The tape curl shall be not greater than 3 ±1/64 inch (76 ±0.4 mm), and the twist shall be not greater than 360 ±5 degrees when tested in accordance with ASTM D3813/D3813M. The tape shall not curl or twist back on to itself.

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3.10 Adhesion at elevated temperature. The tape slippage shall be not greater than 0.125 inch (3 mm) per hour when tested in accordance with ASTM D3654/D3654M, as modified in 5.3.3.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation FAR, or the sustainable acquisition of EO 13693; as applicable.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID. The company furnishing the tape must have been producing a product meeting the requirements of this commercial item description for at least 6 months.

5.3 Testing. All tests shall be conducted, and rolls shall be conditioned prior to testing for a period of 24 hours at a temperature of 73 °F ±3.0 °F (22.8 °C ±1.65 °C) and a relative humidity of 50 ±5 percent.

5.3.1 Dielectric breakdown test. Dielectric breakdown shall be conducted on the primed, etched, or other surface of treated backing material without the adhesive mass. The apparatus shall be in accordance with ASTM D149, utilizing the short-time test with air as the medium. The transformer and source of supply of energy rating shall be not less than 2kVA. The frequency shall be not greater than 100 Hz. The electrodes shall be opposing cylindrical brass rods 1/4 inch (6.4 mm) in diameter with edges rounded to a radius of 1/32 inch (0.8 mm). The upper movable electrode shall weigh 0.1 ±0.005 pound (0.045 ±0.002 kg). If necessary, a special testing device shall be used to clamp the test specimen under pressure in order to prevent flashover around the edges of the material. The voltage shall be increased at a rate of 500 volts/sec.

5.3.2 Varnish adhesion (type II only). An insulation varnish shall be applied to a 6-inch (152-mm) length of the treated backing to a thickness of 4 mils (0.100 mm) and dried for 72 hours. After drying, the tape shall be examined for uniformity of the varnish film and complete surface wetting. The tape shall then be bent 180 degrees over a 1/8 inch (3 mm)

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mandrel. The specimen, after bending, shall be examined for cracking, or peeling away of insulation varnish from backing as specified in 3.3.

5.3.3 Adhesion at elevated temperature. The panel shall be placed in an oven, a 3.53-ounce (100 gm) load shall be uniformly applied to the free end of the tape, and the oven temperature raised to 500 °F (260 °C) in approximately 1 hour. After 1 hour at 500 °F, the specimen shall be examined for slippage. The slippage shall be not greater than 0.125 inch (3 mm). Three specimens shall be tested. Failure of more than one specimen to meet the requirement shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

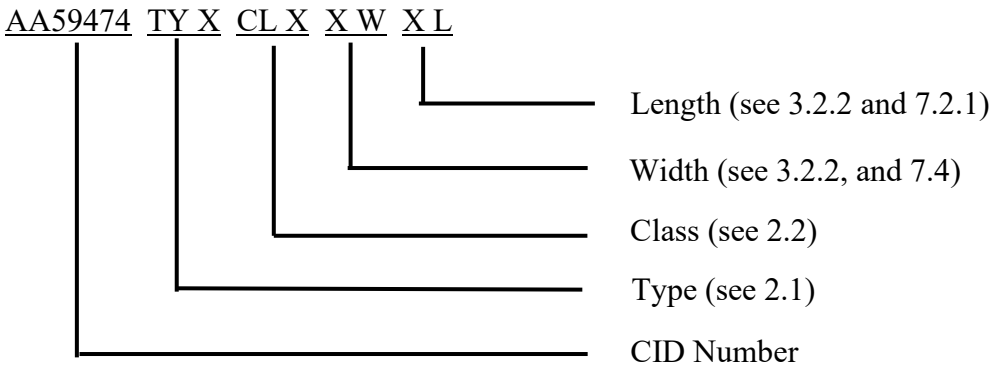
7.1 Intended use. The pressure-sensitive tapes covered by this commercial item description are for electrical systems of underwater ordnance equipment, aircraft, missiles, and other weapons systems within the limitations of the performance requirements.

7.1.1 Type I. Type I tape is intended for splicing insulation of polytetrafluoroethylene-coated electrical wire for continuous operation at 500 °F (260 °C) and where a minimum of 15 lbs/inch (2.63 N/mm) of tensile strength is required for watertight seals around blasting cap and detonating cord, lap splicing of firing cable to prevent shorting out of firing circuit watertight seals around junctions of non-elastic caps and safety fuses, and general use in underwater demolition.

7.1.2 Type II. Type II tape is intended for use in electrical equipment where it is necessary to impregnate with insulation varnish after application of the tape.

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7.2 Part Identification Number (PIN). The PIN used for electrical insulation tape using this CID is generated as follows:



Example: AA59474 TY I CL 6 2.0 W = Type I, Class 6 insulation tape that is 2.0 inches wide and 36 yards long. No length is given in the PIN because the length of the tape is 36 yards long (see 7.2.1).

7.2.1 Length. The length of the insulation tape should only be included in the PIN if the length being procured is not 36 yards.

7.3 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced, if required.
c. Type and class (see 2.1 and 2.2).
d. Backing thickness (see 3.1.1).
e. Core nominal inner diameter (see 3.2.1).
f. The PIN of the insulation tape (see 7.2).
g. Length (if other than 36 yards) and width of tape in inches (see 3.2.2).
h. If marking of the roll is required (3.2.3).
i. The number of splices allowed (if the length is greater than 36 yards) (see 3.2.4).
j. The shelf life (see 3.4).

7.4 Tape widths. Tape widths of 0.5, 1.0, 1.5, and 2.0 inches (13, 25, 38 and 51 mm) are commonly available and used.

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7.5 Source of documents.

7.5.1 ASTM D149, ASTM D257, ASTM D570, ASTM D792, ASTM D3330/D3330M, ASTM D3652/D3652M, ASTM D3654/D3654M, ASTM D3759/D3759M, ASTM D3813/D3813M, and ASTM D4894 are available online at https://www.astm.org.

7.5.2 EO 13693 is available online at https://federalregister.gov.

7.5.3 The Federal Acquisition Regulation (FAR) is available online at https://ecfr.gov.

7.6 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

7.7 Key words

Adhesives
Melting point
Smooth backing
Specific gravity
Treated backing

CONCLUDING MATERIAL

MILITARY INTERESTS:
Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - MI, SM
Navy - EC, OS
Air Force - 20
DLA - GS
Other - DS

Civil Agency Coordinating Activity:
GSA/FAS

Preparing activity:
Navy - AS
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